



## RECENZJA PRACY DYPLOMOWEJ / DIPLOMA THESIS REVIEW

|                                                      |                                                                                                                                              |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| AUTOR / AUTHOR:                                      | Krzysztof Wojdalski                                                                                                                          |
| NUMER ALBUMU / STUDENT'S ID NO.:                     | 310284                                                                                                                                       |
| TYTUŁ PRACY / THESIS TITLE:                          | Microstructure-Aware Deep Reinforcement Learning for High-Frequency Trading                                                                  |
| TYP PRACY / THESIS TYPE:                             | <input type="checkbox"/> PRACA LICENCJACKA / UNDERGRADUATE THESIS<br><input checked="" type="checkbox"/> PRACA MAGISTERSKA / GRADUATE THESIS |
| PROMOTOR / SUPERVISOR:                               | dr Paweł Sakowski                                                                                                                            |
| RECENZENT / REVIEWER:                                |                                                                                                                                              |
| RECENZJA PRZYGOTOWANA PRZEZ /<br>REVIEW PREPARED BY: | <input checked="" type="checkbox"/> PROMOTOR / SUPERVISOR<br><input type="checkbox"/> RECENZENT / REVIEWER                                   |

### WYMAGANIA ODNOŚNIE TRADYCYJNYCH PRAC DYPLOMOWYCH

#### Prace licencjackie

1. muszą mieć jasno określony cel badawczy oraz powinny zawierać pytanie badawcze lub hipotezę badawczą,
2. muszą wykorzystywać aktualną literaturę naukową krajową lub zagraniczną,
3. muszą wykazywać przygotowanie dyplomanta do prowadzenia badań naukowych,
4. muszą wykorzystywać instrumentarium badawcze prezentowane podczas studiów lub wykraczające poza program,
5. mogą mieć formę pogłębionego referatu (o charakterze opisowym), opracowanego na podstawie literatury lub stanowić prezentację badania własnego, przy czym, jeśli praca oparta jest na literaturze przedmiotu, od autora oczekuje się samodzielności konstrukcji i sposobu interpretacji przedstawionego materiału,
6. w przypadku pracy tradycyjnej, powinny mieć objętość do ok. 50 stron standardowego maszynopisu (100 tys. znaków ze spacjami).

#### Prace magisterskie

1. muszą mieć jasno określony cel badawczy oraz zawierać (adekwatnie do obranej metody) hipotezę badawczą lub pytanie badawcze,
2. muszą wykorzystywać aktualną literaturę naukową zagraniczną, w odniesieniu do teorii, metody, jak i wyników empirycznych,
3. muszą wykazywać umiejętność samodzielnego prowadzenia badań naukowych przez dyplomanta,
4. muszą wykorzystywać instrumentarium badawcze prezentowane podczas studiów lub wykraczające poza program,
5. muszą stanowić prezentację badania własnego,
6. w przypadku pracy tradycyjnej, powinny mieć objętość do ok. 80 stron standardowego maszynopisu (160 tys. znaków ze spacjami).

### REQUIREMENTS FOR TRADITIONAL THESES

#### Undergraduate theses

1. must have a clearly defined research objective and should include a research question or a research hypothesis,
2. must make use of current national or international scientific literature,
3. must demonstrate the graduate's preparation for research,
4. must make use of research instrumentation presented during the studies or beyond the programme,
5. may take the form of an in-depth paper (of a descriptive character), developed on the basis of literature, or constitute a presentation of own research; if the paper is based on a given literature, the author is expected to be independent in the construction and interpretation of the presented material,
6. traditional theses should be up to approximately 50 pages of standard typescript (100,000 characters with spaces).

#### Graduate theses

1. must have a clearly defined research objective and include (as appropriate to the method chosen) a research hypothesis or a research question,
2. must make use of current foreign scientific literature, with regard to theory, method as well as empirical results,
3. must demonstrate the graduate's ability to conduct research independently,
4. must make use of research instrumentation presented during studies or beyond the programme,
5. must constitute a presentation of own research,
6. traditional theses should be up to approximately 80 pages of standard typescript (160,000 characters with spaces).

**I. MERYTORYCZNA OCENA PRACY / ASSESSMENT OF THE THESIS CONTENT**(minimum kwalifikujące / *qualifying minimum*: 23pkt.)

| PYTANIE / QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PUNKTY POINTS        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1. MOTYWACJA: Jaki cel (problem, zadanie) został postawiony przed pracą? Czy jego znaczenie zostało uzasadnione?<br>1. MOTIVATION: <i>What goal (task or research problem) has been set in the thesis? Has its significance been justified?</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0 – 5                |
| The thesis sets a clearly defined research problem: whether a continuous-control deep reinforcement learning agent, conditioned on limit-order-book microstructure features, can extract a risk-adjusted edge at tick frequency, and what economically bounds that edge. The significance is convincingly argued on two levels - theoretically, because at tick frequency execution conditions are part of the signal environment rather than an implementation detail, so the decision problem genuinely differs from bar-based trading; and methodologically, because the applied RL-trading literature is shown to be weak precisely on cost modelling and evaluation discipline. The author explicitly frames the study as a controlled simulation rather than a live HFT system and states from the outset what is not modelled (latency, queue priority, partial fills, market impact), which is the right way to establish scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5                    |
| 2. PYTANIE BADAWCZE LUB HIPOTEZA BADAWCZA: Czy w pracy zostały postawione pytania badawcze i/lub hipotezy badawcze? Jakie? [Wymogi w tym zakresie zostały podsumowane na stronie 1 formularza recenzji.]<br>2. RESEARCH QUESTION OR HYPOTHESIS: Have research <i>questions and/or research hypotheses</i> been posed in the thesis? If so, which ones? [Please note that the requirements in this matter are summarized on the first page of the review form.]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0 – 5                |
| Four numbered hypotheses are stated in Section 1.1.2 and carried consistently through the entire work: (H1) a TD3 agent trained on LOB features learns a risk-adjusted edge that survives out of sample; (H2) that edge depends on the assumed transaction cost; (H3) a broader microstructure-aware state representation improves out-of-sample performance; (H4) the reward objective shapes the policy and not only its score. Each is designed as a single-factor comparison, in which scenarios differ only in the quantity under study, which is exactly what makes the hypotheses testable rather than merely illustrative. The author also frames significance against an economic threshold rather than against zero, correctly observing that with roughly a million steps per instrument any statistical test rejects at effect sizes far below a tradable margin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5                    |
| 3. METODA: Jaką metodą badawczą posłużono się w pracy? Czy jest ona dostosowana do celu pracy? Czy została prawidłowo zastosowana?<br>a) wybór metody i dostosowanie do celu pracy, b) zakres i poprawność użycia.<br>3. METHOD: <i>What research method was used in the thesis? Does it fit the goal of the thesis? Has it been applied properly?</i><br>a) <i>choice of method and its fit to the goal of thesis, b) scope and adequacy of use.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | a) 0 – 5<br>b) 0 – 5 |
| a) Choice of method and fit to the goal (5/5). TD3 is a well-motivated choice: the exposure decision is naturally continuous, and Chapter III derives the progression from value-based methods through DDPG and PPO to TD3's twin-critic and target-smoothing corrections, so the selection is argued rather than assumed. Training DDPG and PPO under an identical configuration is a deliberate and correct design move. It converts H1 from a claim about one optimiser into a claim about a learnable signal. The evaluation apparatus (buy-and-hold, TWAP, VWAP and a random-action control, chronological splits, strictly causal online normalisation, one varying factor per hypothesis) fits the goal well.<br><br>b) Scope and adequacy of use (3/5). The implementation itself is careful and well documented, but the empirical scope is narrow and this materially limits what the apparatus can deliver. The dataset covers only four trading days (25-27 February 2026 for training, 2 March 2026 split into validation and test), so the agent is never exposed to a different volatility or spread regime, each configuration is trained on a single seed, and three of the reported runs failed to train at all, collapsing to a constant saturated position, which leaves H3 and H4 unresolved. In addition, all headline results are produced under frictionless mid-price execution, and the bid/ask repricing that tests their robustness is applied post hoc to a single instrument without retraining. The author identifies every one of these limitations explicitly and diagnoses the saturation failure carefully, which is to his considerable credit, but they nevertheless constrain the adequacy of the empirical application. Against this, the complete code base is published as an open repository and is treated as an integral part of the work rather than as supporting material: experiments are specified declaratively, so each hypothesis scenario can be re-run from its configuration file, which makes the causal normalisation, the chronological splits and the one-varying-factor design auditable rather than merely asserted. | a) 5<br>b) 3         |
| 4. WIEDZA: Czy i jaką wiedzę nabytą w toku studiów danego stopnia na studiowanym kierunku wykorzystano w pracy? Jak można ocenić stopień przyswojenia wiedzy (podstawowy, zadowalający,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0 – 5                |

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| <p>pogłębiony, rozszerzony)? Czy student potrafi łączyć wiedzę z różnych obszarów, korzystać z dorobku różnych dziedzin (jakich?) itd.</p> <p>4. <i>KNOWLEDGE: If and what kind of knowledge gained during the given degree studied in the field of studies has been used in the thesis? How to rate the level of acquired knowledge (basic, satisfactory, advanced, deepened)? Has student been able to link alternative sources of knowledge, benefit from other disciplines (which ones) etc.?</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |
| <p>The thesis makes substantial and correct use of knowledge from the Quantitative Finance programme and beyond it. Market microstructure theory is used at a deepened level - Glosten and Milgrom, Kyle, Roll, Harris and the spread decomposition are not merely cited but mapped one-to-one onto named state variables, so that each theoretical mechanism yields an observable proxy together with a statement of its limitation. Reinforcement learning is developed from first principles through to the specific instabilities of deterministic actor-critic methods, and the work also draws on software engineering (a declaratively configured, reproducible experiment pipeline) and on the reproducibility literature in deep RL (Henderson et al., 2018). The level of acquired knowledge is deepened, and the linkage across disciplines is genuine.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>5</b>                   |
| <p>5. <i>LITERATURA I DANE: Czy dobór literatury i/lub materiału empirycznego jest trafny i aktualny?</i><br/> a) dobór literatury i/lub materiału empirycznego, b) wykorzystanie ich w pracy.<br/> 5. <i>LITERATURE AND DATA: Is the selection of literature and/or empirical material accurate and relevant?</i><br/> a) selection of literature and/or empirical material, b) usage in the thesis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | a) 0 – 5<br>b) 0 – 5       |
| <p>a) Selection of literature and empirical material (5/5). The bibliography comprises roughly 90 items and is dominated by current international sources spanning theoretical microstructure, deep reinforcement learning, and applied RL trading, with attention to reproducibility and evaluation-rigour work. The empirical material is exactly right for the question: Nasdaq MBP-10 (ten-level market-by-price) data obtained through DataBento for six liquid US equities, which is the granularity actually required to construct the queue, imbalance and book-curvature features the design relies on. The only reservation concerns the length of the sample rather than its quality, and is scored under question 3b.</p> <p>b) Usage in the thesis (5/5). Chapter II is not a survey appended for form's sake. It is structured to deliver the design decisions of Chapter IV, separating descriptive microstructure variables from tradable signals by an explicit criterion (available before the action, survives costs, improves out-of-sample performance). Prior applied RL work is used to structure the treatment of action-space design, transaction-cost modelling and validation rigour. The data are also used properly: the per-instrument decomposition in Table 5 shows that cumulative return across instruments largely reflects event frequency rather than skill, and the author correctly switches to a per-step measure.</p> | a) <b>5</b><br>b) <b>5</b> |
| <p>6. <i>WYNIKI: Czy został zrealizowany cel pracy? Czy zweryfikowano hipotezy badawcze? Czy uzyskano odpowiedź na pytania badawcze? Czy praca zawiera błędy, które rzutują na poprawność wyciągniętych wniosków?</i><br/> 6. <i>RESULTS: Has the aim of the thesis been achieved? Have the research hypotheses been verified? Have research questions been answered? Does the thesis have any mistakes which cast on the correctness of driven conclusions?</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>0 – 5</b>               |
| <p>The aim is partly achieved and the verdicts are stated with unusual precision. H1 is supported in a qualified form - all three learners separate cleanly from the random control while remaining indistinguishable from one another, so a learnable signal is present, but the per-step edge is smaller than a half-spread and reverses sign when the action path is repriced at the bid/ask. H2 is partially supported: the policies are cost-sensitive but no break-even fee is identified. H3 and H4 are explicitly reported as unresolved, because in each comparison one or two runs failed to train. What prevents this from being simply a shortfall is the author's own synthesis: three separately configured zero-fee runs collapsed to a saturated position under identical architecture and budget, which locates the binding constraint in training stability rather than in feature or reward design, and this is presented as the study's most substantial finding. No error was found that casts doubt on the conclusions drawn; an earlier evaluation-window mismatch in the H1 comparison was detected, re-measured over the full test split and disclosed in Section 7.2.4.</p>                                                                                                                                                                                                                                                          | <b>4</b>                   |
| <p>7. <i>LOGIKA WYWODU: Czy sposób prowadzenia i opisu wywodu jest prawidłowy? Czy zachowana została logika wywodu? Czy dobrze poprowadzona została narracja?</i><br/> 7. <i>LOGIC OF THESIS: Is the way of conducting and describing arguments correct? Has the logic of arguments been preserved? Was the narrative well conducted?</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>0 – 5</b>               |
| <p>The argument is conducted rigorously and the logic is preserved throughout: motivation, microstructure evidence, RL foundations, agent design, implementation, hypothesis-by-hypothesis results, then a synthesis that separates what is established from what is not. In my opinion, the narrative discipline is the strongest feature of the thesis. The author distinguishes carefully between a failure that is an economically rational response to fees and a failure that is an optimisation artefact. Chapter VII does not retreat into generalities but converts the limitations into a prioritised research plan.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>5</b>                   |
| <b>Suma punktów / Total points:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>42 / 45</b>             |

## II. FORMALNA OCENA PRACY / FORMAL ASSESSMENT OF THE THESIS

(minimum kwalifikujące / qualifying minimum: 7pkt.)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |
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| 1. TYTUŁY: Czy poprawnie i adekwatnie do treści sformułowano tytuł pracy i jej rozdziałów / podrozdziałów?<br>1. TITLES: Are the title of the thesis and its subchapters formulated correctly and adequately to its content?                                                                                                                                                                                                                                       | 0 - 2          |
| The title accurately and economically describes the content: microstructure-aware state design, deep reinforcement learning, and high-frequency trading are all genuinely central to the work. Chapter and section titles are informative and adequate. The empirical chapter is organised by hypothesis, so the reader can navigate directly from a claim in the introduction to the evidence that bears on it.                                                   | 2              |
| 2. KONSTRUKCJA: Czy konstrukcja pracy jest właściwa?<br>2. CONSTRUCTION: Is the project structured correctly?                                                                                                                                                                                                                                                                                                                                                      | 0 - 2          |
| The logical construction is sound. The deduction concerns proportion and length: the thesis runs to 157 pages, with a main body of roughly 110 pages against the guideline of approximately 80 pages of standard typescript for a master's thesis. Chapters III and IV are long relative to the empirical payload, and part of the expository RL material and of the implementation detail could have been moved to the appendices without any loss.               | 1              |
| 3: WSTĘP I ZAKOŃCZENIE: Czy wstęp i zakończenie dobrze informują o zamierzeniach Autora(ki), zawartości pracy, użytej metodzie i osiągniętych rezultatach?<br>3. INTRODUCTION AND CONCLUSIONS: Does the introduction and the concluding chapter clearly state the Author's intentions, thesis contents, the method used and the results achieved?                                                                                                                  | 0 - 2          |
| The Introduction states the research problem, the claimed novelty, the four hypotheses, the objectives, the chapter plan, and the role of the accompanying repository. Chapter VII summarises the evidence hypothesis by hypothesis, states plainly which claims are established and which are not, sets out the limitations under four organised headings, and derives the implications for trading-system design. Both fulfil their function fully and honestly. | 2              |
| 4. ODWOŁANIA: Czy technika prezentacji cudzego dorobku jest poprawna (odsylacze, cytaty, omówienia, bibliografia itp.)?<br>4. REFERENCES: Are the referencing and quotation techniques applied correctly (references, quotes, discussions, bibliography etc.)?                                                                                                                                                                                                     | 0 - 2          |
| Referencing follows a consistent APA-style convention with DOIs and stable links, and the bibliography is complete. Attribution is precise: prior results are discussed rather than merely listed, and the boundary between what the literature establishes and what this thesis contributes is stated explicitly at the end of Chapter II. Data sources and the vendor schema are cited properly, including the MBP-10 schema documentation.                      | 2              |
| 5. EDYCJA: Czy właściwie skonstruowano tabele, wykresy, aneksy statystyczne itp.? Czy edycja nie budzi zastrzeżeń?<br>5. EDITION: Are charts, graphs, and statistical appendices built appropriately? Is their graphic form acceptable?                                                                                                                                                                                                                            | 0 - 2          |
| Tables are well constructed, each with an explicit note stating the execution assumption under which the figures hold, a legend defining every metric and a source line. Figures, the pipeline diagram and the appendix material are appropriate and legible. The editorial standard is high throughout.                                                                                                                                                           | 2              |
| 6. JEZYK: Czy praca została napisana zwięźle, poprawnie z punktu widzenia literackiego, itp.?<br>6. LANGUAGE: Was the thesis written concisely, correctly from a literary point of view, etc.?                                                                                                                                                                                                                                                                     | 0 - 2          |
| The thesis is written in correct, concise and often elegant English. No significant grammatical or stylistic problems were identified.                                                                                                                                                                                                                                                                                                                             | 2              |
| 7. TERMINOLOGIA: Czy praca została napisana poprawnie pod względem terminologicznym? Czy ewentualne błędy terminologiczne w pracy rzutują na poprawność wyciągniętych wniosków?<br>7. TERMINOLOGY: Was the thesis written correctly in terms of terminology? Do any terminological errors in the thesis affect the correctness of the conclusions?                                                                                                                 | 0 - 2          |
| Terminology is used correctly and consistently, both in the microstructure and in the reinforcement-learning parts, and terms that carry specific meanings are defined before use and applied consistently. The glossary and the notation appendix support this. No terminological error was found that would affect the correctness of the conclusions.                                                                                                           | 2              |
| <b>Suma punktów / Total points:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>13 / 14</b> |

## III. INNE UWAGI / ADDITIONAL COMMENTS

|                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Zakres ewentualnych poprawek pracy (jeśli praca ich wymaga).                                                                                                                                                                                                                                                                                               |
| 1. <i>Scope of thesis corrections (if required).</i>                                                                                                                                                                                                                                                                                                          |
| NA                                                                                                                                                                                                                                                                                                                                                            |
| 2. Czy praca została napisana w formie artykułu naukowego? TAK/NIE                                                                                                                                                                                                                                                                                            |
| 2. <i>Information if the thesis was written in the form of scientific paper. YES/NO</i>                                                                                                                                                                                                                                                                       |
| <input type="checkbox"/> TAK / YES                                                                                                                                                                                                                                                                                                                            |
| <input checked="" type="checkbox"/> NIE / NO - the thesis has the traditional chapter-based form.                                                                                                                                                                                                                                                             |
| 3. Czy współautorstwo pracy jest uzasadnione (jeśli praca jest pisana przez dwóch studentów). TAK/NIE                                                                                                                                                                                                                                                         |
| 3. <i>Information if co-authorship is justified (if the thesis was written by two students). YES/NO</i>                                                                                                                                                                                                                                                       |
| <input type="checkbox"/> TAK / YES                                                                                                                                                                                                                                                                                                                            |
| <input checked="" type="checkbox"/> NIE / NO - not applicable, the thesis has a single author.                                                                                                                                                                                                                                                                |
| 4. Konkursy                                                                                                                                                                                                                                                                                                                                                   |
| Czy praca powinna zostać zgłoszona do konkursu na najlepszą pracę licencjacką im. W. Baki (jeśli dotyczy)? TAK/NIE                                                                                                                                                                                                                                            |
| Czy praca powinna zostać zgłoszona do konkursu na najlepszą pracę magisterską im. A. Semkowa (jeśli dotyczy)? TAK/NIE                                                                                                                                                                                                                                         |
| 4. <i>Contests</i>                                                                                                                                                                                                                                                                                                                                            |
| <i>Should the thesis be submitted to the W. Baka competition for the best undergraduate thesis (if applicable)? YES/NO</i>                                                                                                                                                                                                                                    |
| <i>Should the thesis be submitted to the A. Semkow competition for the best graduate thesis (if applicable)? YES/NO</i>                                                                                                                                                                                                                                       |
| <input checked="" type="checkbox"/> TAK / YES - I recommend submitting the thesis to the competition for the best master's thesis. The single-factor experimental design, the treatment of the code repository as an auditable part of the work, and the willingness to report and diagnose failed runs rather than suppress them make it a strong candidate. |
| <input type="checkbox"/> NIE / NO                                                                                                                                                                                                                                                                                                                             |

#### IV. OCENA PRACY / GRADING SCALE

- ocena merytoryczna / *substantive assessment* 0 – 45 pkt. (min. kwalifikujące / *min. passing score*: ~50%, 23 pkt.);
- ocena formalna / *formal assessment* 0 – 14 pkt. (min. kwalifikujące / *min. passing score*: ~50%, 7 pkt.);
- ocena łączna / *total score* 0 – 59 pkt.

| Ocena pracy / <i>Thesis grade</i>                                                 | Próg procentowy / <i>Percentage thresholds</i> | Liczba punktów / <i>Points</i> |
|-----------------------------------------------------------------------------------|------------------------------------------------|--------------------------------|
| Ocena celująca / <i>Excellent (5!)</i>                                            | do / <i>up to</i> ~ 100%                       | 57-59                          |
| Ocena bardzo dobra / <i>Very good (5)</i>                                         | do / <i>up to</i> ~ 95%                        | 48-56                          |
| Ocena dobra i dobra+ / <i>Good and good plus (4, 4+)</i>                          | do / <i>up to</i> ~ 80%                        | 40-47                          |
| Ocena dostateczna i dostateczna+ / <i>Satisfactory, satisfactory plus (3, 3+)</i> | do / <i>up to</i> ~ 65%                        | 30-39                          |
| Ocena niedostateczna / <i>Unsatisfactory (2)</i>                                  | do / <i>up to</i> ~ 50%                        | 0-29                           |

**Praca wybitna (ocena 5!)** – praca znacząco wykraczająca poza program studiów, zawierająca istotny wkład w naukę, nadająca się do publikacji w uznanych czasopismach międzynarodowych.

**Outstanding work (grade 5!)** - a thesis which significantly exceeds the program of the studies, contains essential input to the science and can be published in recognised scientific journals.

**Praca bardzo dobra (ocena 5)** – praca w znaczący sposób wykorzystująca treści określone w programie kształcenia, w pełni spełniająca wymagania stawiane pracom dyplomowym.

**Very good work (grade 5)** - a thesis which significantly uses content specified in the curriculum, which fully meets the general requirements for the thesis.

**Praca dobra (ocena 4)** – praca w umiarkowany sposób wykorzystująca treści określone w programie kształcenia, w przeważającej części spełniająca wymagania stawiane pracom dyplomowym.

**Good work (grade 4)** - a thesis which uses content specified in the curriculum in a moderate way and meets the general requirements for thesis in major part.

**Praca słaba (ocena 3)** – praca w dostateczny sposób wykorzystująca treści określone w programie kształcenia, połowicznie spełniająca wymagania stawiane pracom dyplomowym.

**Satisfactory work (grade 3)** - thesis sufficiently uses content specified in the curriculum, partly meets the general requirements for the thesis.

**Praca zła (ocena 2)** – praca w niedostateczny sposób wykorzystująca treści określone w programie kształcenia, w niewielkim stopniu spełniająca wymagania stawiane pracom dyplomowym, zawierająca istotne błędy merytoryczne wpływające na wyciągnięcie niepoprawnych wniosków.

**Unsatisfactory work (grade 2)** - thesis insufficiently uses content specified in the curriculum, meets the general requirements for thesis in a small extent, contains significant substantive errors which effect in incorrect conclusions.

|                                                                                                                                                                            |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>I. MERYTORYCZNA OCENA PRACY / ASSESSMENT OF THE THESIS CONTENT</b><br>Suma punktów / <i>total points</i> : (minimum kwalifikujące / <i>qualifying minimum</i> : 23pkt.) | <b>42</b> |
| <b>II. FORMALNA OCENA PRACY / FORMAL ASSESSMENT OF THE THESIS</b><br>Suma punktów / <i>total points</i> : (minimum kwalifikujące / <i>qualifying minimum</i> : 7pkt.)      | <b>13</b> |
| <b>Suma punktów / Total points:</b>                                                                                                                                        | <b>55</b> |

**OCENA / GRADE:** 5 (bardzo dobra / very good)

Warszawa, .....  
(Warsaw, data / date)



Podpisany elektronicznie przez  
Paweł Marcin Sakowski  
10.09.2026  
10:02:38 +02'00'

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(Imię i nazwisko / Name and surname, PODPIS / SIGNATURE)